

PROFESSIONAL SUMMARY:

Results-driven **Senior Software Engineer (Infrastructure & Platform)** with over **8+ years** of industry experience in designing and securing large-scale workloads, multi-cloud infrastructures, microservices architectures and **developer productivity**, with **golden paths**. Proficient in **Python, Golang, Kubernetes, Terraform, AWS, Azure, GCP, CI/CD, Infrastructure Security**, and **Containerization**. I architect robust **Kubernetes-based security controls** (OPA/Gatekeeper, IAM/RBAC, Network Policies), lead external **HIPAA, PCI-DSS** and **CCF** audits, and build **SBOM** pipelines that eliminate high-risk dependencies. Passionate about **developer productivity**, I've optimized build and dev cycles by ~70%, created self-service **CI/CD** and **GitOps** tooling for **80+ teams**, and integrated **observability** and **GenAI**-driven code-assist solutions to accelerate and secure the entire software lifecycle.

CORE SKILLS:

Platform Engineering & SRE: K8s, EKS, AKS, GKE, OpenShift/ROSA, Cilium, Karpenter, CoreDNS, Helm, Argo CD, GitOps, GitHub Runners, GitHub Actions, golden paths, SLIs/SLOs, on-call, incident response, postmortems, **gameday/Fire-Drill**, runbooks, production readiness.

Cloud & Infrastructure as Code: AWS, Azure, GCP, **Terraform**, OpenTofu, **Terragrunt**, Atlantis, Crossplane, CloudFormation, Ansible, multi-account AWS, IAM, IRSA, KMS, SNS, SQS, SES, RDS, Lambda, App Runner, CloudTrail.

DevSecOps & Compliance: Kubernetes security, OPA/Gatekeeper, IAM/RBAC, network policies, container scanning, SBOM pipelines, cosign image signing, Wiz, SAST/SCA concepts, vulnerability remediation, HIPAA, PCI-DSS, CCF, audit evidence, secure SDLC

Observability & AIOps: Datadog, Prometheus, Grafana, Thanos, ELK, Fluent Bit, OpenTelemetry, LangSmith, alerting, telemetry pipelines, AWS Health automation, Slack-native triage, AI-assisted incident workflows.

Programming & Automation: Python, Go, Bash/Shell, Rego, FastAPI, Django, Flask, REST API's, distributed systems debugging, Kubernetes informer/cache patterns, concurrency, automation tooling

AI Infrastructure & Agentic Workflows: LangGraph, MCP, Temporal, Claude, OpenAI, Gemini, multi-agent SRE workflows, tool integrations, evaluation traces, approval-gated infrastructure campaigns.

CERTIFICATIONS:

Cisco Certified Network Associate (**CCNA CSCO13315295**), Certified Kubernetes Administrator (**CKA, LF-v9d9mtx4py**), Certified Kubernetes Security Specialist (**CKS**)

PROFESSIONAL EXPERIENCE:

Senior Platform Engineer (Frictionless & AI Ops), Circle Internet Financial, CA, USA **August 2025 – Present**

- Delivered a **fleet-wide** Terraform EKS module migration/jump from **v17 to v20** across 90+ Kubernetes clusters and 40+ AWS accounts, resolving 10 structural dependency cycles and authoring 20 state-migration blocks for a **zero-downtime rollout**.
- Operate and maintain 90+ EKS clusters with Cilium CNI, Fluent Bit, ArgoCD, Karpenter, CoreDNS, and Datadog, supporting multi-region production infrastructure and regulated financial workloads.
- Diagnosed chronic **production node churn** by proving auto-scaler misconfiguration was converting **On-Demand capacity to Spot**, architected a hybrid capacity model with disruption budgets that **reduced churn by 85%** and stabilized cluster networking health.

- Led **production security incident forensics** by correlating IdP logs and cloud telemetry to distinguish automated API activity from human action, then **drove cross-functional** remediation with Security, Identity, IT, and SRE leadership.
- Built reusable Terraform modules for **cross-account AWS SNS/SQS/KMS** publishing and SES email infrastructure, automating KMS CMK creation, IAM policy granularity, SMTP credentials, DKIM/SPF, and Proofpoint alignment for secure self-service adoption.
- Diagnosed and **resolved** recurring DaemonSet OOMs across CI clusters by profiling Kubernetes informer cache behavior from **GitHub runner** workloads, implemented event-driven caching, informer filtering, bounded concurrency, and **Go runtime memory tuning**.
- Participate in **on-call rotations** for production infrastructure, triaging SLO breaches, CNI failures, metrics pipeline gaps, and cluster-level incidents, write structured incident reports and drive follow-up remediation to closure.
- Built an event-driven alerting and **auto-ticketing pipeline for AWS Health events** with tiered severity routing, noise suppression, and automatic Jira ticket creation for production-impacting scheduled changes.
- Strengthened **incident readiness through biannual game-day** exercises, operational runbooks, node health automation, Node Monitoring Agent enablement, and Fluent Bit **configuration consolidation** via Python tooling.
- Delivered Scaffold, a **4-agent AI pipeline** for safe Terraform module migrations that classifies plan deltas, subtracts baseline drift, and generates migration artifacts to eliminate destructive plan churn across 90+ clusters.
- **Tech stack:** Python, Go, Bash, AWS, GCP Vertex AI, Kubernetes, Terraform, OpenTofu, Terragrunt, Atlantis, Helm, ArgoCD, Cilium, Karpenter, Fluent Bit, Datadog, OpenTelemetry, LangSmith, Okta, Cloudflare, GitOps, Slack, LangGraph, MCP, Claude, OpenAI, Gemini, Agent-Skills.

Senior Software Engineer, Adobe Inc, San Jose, California, USA

February 2021 – June 2025

- **Led a team of 6 platform security-ops** engineers through roadmap execution, **risk management**, delivery metrics, and platform resilience initiatives across platform and security engineering organizations.
- Directed critical **initiatives across teams and organizations** of platform & security engineers, collaborating closely with product and program managers to shape **strategic roadmap priorities**, drive timely execution, and ensure platform capabilities met key business objectives.
- Architected, implemented, and maintained **secure Kubernetes**-based infrastructure across AWS, Azure, OpenShift, and ROSA, using **automation-first** patterns and security-by-default controls.
- Developed and enforced **robust security policies** using **OPA/Gatekeeper**, IAM/RBAC, and network policies across globally distributed public and private cloud environments.
- Integrated **Wiz CLI** into CI pipelines for automated container vulnerability scanning and implemented cryptographic **image signing with cosign** to strengthen supply-chain integrity.
- Built automated **SBOM-generation** pipelines, artifact-validation gates, and code-hygiene checks that eliminated 75% of high-risk transitive dependencies.
- Built and maintained CI/CD pipelines using Jenkins, **ArgoCD**, **Helm**, and **GitOps** practices to improve deployment consistency and reduce operational risk.
- Integrated **observability** and telemetry into CI/CD and production workflows using Prometheus, Grafana, and ELK, surfacing actionable metrics for performance tuning and reliability improvement.
- Automated environment provisioning, patch management, **infrastructure-as-code** workflows, and incident remediation at scale with **Terraform**, Ansible, and Python, cutting manual toil by 60%.

- Partnered with **Adobe Compliance and external auditors** to achieve and maintain HIPAA, PCI-DSS, and CCF attestations, led **threat modeling**, vulnerability assessments, **penetration testing** coordination, and remediation of audit findings.
- **Tech Stack:** Python, Go, ReGo, AWS, Azure, Kubernetes, Docker, Terraform, CrossPlane, CloudFormation, Ansible, Jenkins, ArgoCD, Prometheus, Grafana, ELK, CI/CD, GitOps, Helm.

Software Engineer – II, ioTium Inc, Santa Clara, CA, USA

October 2018 – January 2021

- Design, develop and implement microservices **architecture** for **containerized** data processing and Network Monitoring on Linux VMs and edge devices with **Kubernetes** for orchestration, VMWare and ELK stack for device management in process control secure environment.
- Design develop and implemented **Simulators** for data generation and data transfer tools to push the simulated data to the remote destinations such as Azure IoT-Hub, AWS Greengrass, Google IoT using **MQTT/AMQP/HTTPS** protocols.
- Build software **automation** workflows for **infrastructure monitoring** and alerting by leveraging Python automation framework.
- **Initiated proposal** for using Industrial IoT as a platform service to provide predictive maintenance, OEE and optimizing spare parts efficiency in the equipment division by conducting customer site surveys and interviews.
- Work together with marketing, **customer success and Engineering** to understand the end user/customer requirements and translate them into solutions.
- Responsible for developing ioTium Edge services using Golang, **Python** and **Ansible** playbooks for deployment automation.
- Worked on Various Industrial IoT devices and Protocols such as **BACnet, Modbus, OPC**, RockWell PLC, Siemens S7-300 PLC
- **Tech Stack:** Python, GoLang, Bash/Shell, C, Kafka, InfluxDB, Docker, Kubernetes, AWS, Azure, GCP, BitBucket, Telit, ELK, **TCP/IP** Suite.

Software Developer (Intern), Octonius Inc, Palo Alto, CA, USA

May 2017 – October 2017

- Responsible for Creating **voice integration** and VOIP Search Functionality.
- Experienced in integrating file transfer-based **Search Engine** along with Voice based search using Web-kit-speech Recognition API.
- Worked on **Machine Learning algorithm** to predict the user-based recommendation based on the pattern analysis.
- Responsible for designing a test framework to validate the code changes before deploying on production servers.
- Worked on **monitoring** the software patch, testing, bug analyzing and providing a fix collaborating with other developers.
- Write REST APIs, helper functions and test automation scripts for the internal **test automation** framework using JIRA and Python.
- Engage with customers and **support** to solve production issues.
- **Tech Stack:** Python, Node.JS, MongoDB, PostgreSQL, CloudFormation, Rest API, Jenkins, Bitbucket, AWS, Docker.

IT Support (Student Assistant), San Jose State University, San Jose, CA, USA

February 2017 – May 2018

- Maintained Cisco routers and switches, configured **L2, L3 protocols** on routers.
- Troubleshooted the routers (routing protocols) and switches (**VLans**, STP), helped Students to resolve technical issues.
- Assist in modelling the site blueprints by taking into consideration the RF parameters and thus deploy the access points.
- Experienced **troubleshooting network problems** for LAN and Wireless communications and security issues.

EDUCATION:

Master of Science (Electrical Engineering - Computer Networks & Deep Learning)
San Jose State University, San Jose, CA, USA

May 2018

Bachelor of Technology (Electronics & Computer Engineering)
VR Siddhartha Engineering College, Affiliated to JNTU'K

June 2016

ACADEMIC EXPERIENCE:

[Slack Bot for coursework info](#) - Slack API, Python, DynamoDB, Quepy, AWS Lambda, NLTK

- Built a question-answer chat bot against data stored in AWS DynamoDB and a Slack API.
- Defined regular expressions to match income natural language questions (Lemmas) using NLTK tagger.
- Configures AWS API gateways and deployed the business logic on AWS Lambda using Python-Lambda libraries.

[Predict Survival on the Titanic](#) – Machine Learning, Python, Jupyter Notebook, Sklearn

- Feature Engineering and data cleaning to handle non-numeric, missing and categorical values.
- Used Support Vector Classifier to train model and predict survival with an accuracy of 76%.

[Develop a Load Balancing Algorithm for SDN using POX controller](#) - Python, SDN, FloodLight Controller

- Lead a team in developing and implementing a load balancing algorithm using FloodLight controller.
- Was involved in debugging and developing the load balancing code.
- Developed and Implemented Weighted Round Robin load balancing, Dijkstra algorithm and was developed using Python.

[URLShorteningService](#) - Python, Django, SQL, Heroku, Phishtank API, Docker, HTML, CSS

- Used Django to make web service calls. Dockerize the application to deploy in Heroku.
- Implemented Memcachier/Memcached to hold previous calls in memory thus increasing the performance when dealing with domain/urls already cached.
- Implemented Indexing for Speedy retrieval, and duplicate checks to make sure URL is stored only one time.

[Stateful Firewall Software](#) - Python, Wireshark, Virtual Environment, IPtables, Scapy, Shell scripting

- Basic Defence mechanism to drop malicious packets using the log file. Aim to deliver packet filtering, IP filtering, blocking Dos attacks and malicious website requests with minimum response time.